Evaluation Report of EliteFS_09_Mar_AttentionAllYouNeed-KMIT

Average Score: 8/10 | Attempted: 5 of 5 questions

Question 1

THEORY REMEMBER Best Score: 10/10

What was the problem in the embedding mechanism of Word2Vec?

Desired Answer

The main limitation of Word2Vec embeddings is that they are **static**, meaning each word is assigned a single fixed vector representation regardless of the context in which it appears. This creates difficulties when words have **multiple meanings (polysemy)**. For example, the word *"bank"* in the phrases *"river bank"* and *"bank loan"* would have the same embedding, even though the meanings are different. Because Word2Vec does not consider context when generating embeddings, it cannot capture context-dependent meanings of words.

Student's Answer

In Word2Vec embedding mechanism, it could not address the semantic meaning of words, like "bank" has different meanings in different context, eg: I went for a walk near the bank of river, refers to river bank whereas: I went to bank to withdraw money, refers to the other bank. As Word2Vec generates only 1 vector embedding per word it becomes difficult to cater the multi-meaning needs of the words.

Evaluations

Feedback 1

★★★★★ 10/10

The student correctly recalls that Word2Vec produces a single, fixed vector per word ('only 1 vector embedding per word'), leading to inability to capture context-dependent meanings (polysemy). They illustrate this with the 'bank' example in both 'river bank' and 'bank loan,' matching the Actual Answer's points about static embeddings, multiple meanings, and lack of contextual consideration. No key factual element from the Actual Answer is missing or incorrect.

Feedback 2 ★★★★★ 10/10

The student correctly recalls that Word2Vec assigns a single fixed vector per word (static embeddings), ignores context, and therefore cannot capture words' multiple meanings (polysemy). They even supply the same "bank" example. All key points from the Actual Answer are accurately and completely reflected.

Question 2

THEORY REMEMBER Best Score: 8/10

What do you mean by Multi-Head Attention?

Desired Answer

Multi-Head Attention is a mechanism used in Transformers where **multiple attention operations are performed in parallel** instead of using a single attention mechanism. Each attention head learns to focus on **different relationships between words in a sentence**, such as syntactic structure, semantic connections, or positional dependencies. After computing attention in each head, the results are combined to form a richer and more informative representation of the sequence. This allows the model to capture multiple types of linguistic patterns simultaneously.

Student's Answer

In multi-head attention, multiple Attention mechanisms are used, each proficient in its work, for example: in a sentence, one head might look at the subject of the sentence, other head at the verbs, other at the meaning of certain words. As each head is working on retrieving relevant information from the text, therefore multi-head attention

Evaluations

Feedback 1 ★★★½☆ 7/10

The student correctly recalls that multi-head attention uses multiple attention mechanisms (heads) in parallel and that each head focuses on different relationships in a sentence (e.g. one on subjects, another on verbs, another on word meanings), which matches the core of the Actual Answer. However, they omit any mention of how the results from each head are combined to form a richer and more informative representation and do not explicitly note positional dependencies or the final fusion step. These missing details prevent a full (9–10)

score but the primary concepts are accurately recalled.

Feedback 2

★★★★☆ 8/10

The student correctly states that multi-head attention uses multiple attention mechanisms in parallel and that each head focuses on different aspects of a sentence (e.g., subject, verbs, word meaning), which corresponds to the Actual Answer's points about multiple parallel attention operations and different syntactic/semantic relationships. However, they omit the key step that the outputs of all heads are combined into a richer, unified representation to capture multiple linguistic patterns simultaneously. This missing detail prevents a perfect score.

Question 3

THEORY REMEMBER Best Score: 6/10

What was the basic problem in LSTM from an NLP perspective?

Desired Answer

Before LSTM, Recurrent Neural Networks (RNNs) were used to model sequences in NLP. However, RNNs suffered from the **vanishing gradient problem**, which made it difficult for them to learn relationships between words that were far apart in a sentence. As a result, they could only remember short-term context and often failed to capture long-term dependencies in language. LSTM was introduced to address this issue by incorporating **memory cells and gating mechanisms** that allow important information to be stored and preserved over long sequences.

Student's Answer

LSTM is Long Short Term memory, the basic problem with LSTM was, as the sentence grew longer, the model no longer could remember the old words or sentences.. Leading to memory loss, and inefficiency..

Evaluations

Feedback 1

★★½☆☆ 5/10

The student correctly identifies that the core issue was the model's inability to remember

information over long sentences (i.e. long-term dependencies), which aligns with the Actual Answer's point that RNNs suffered from vanishing gradients and could only remember short-term context. However, the student fails to mention the vanishing gradient problem by name and omits any reference to how LSTM solves the issue using memory cells and gating mechanisms. Thus, they capture the general memory-loss issue but miss the key terminology and solution details from the Actual Answer.

Feedback 2

★★★☆☆ 6/10

The student correctly recalls the core fact that the model could not remember words in longer sentences (i.e., inability to capture long-term dependencies), which aligns with the Actual Answer's point that standard RNNs suffer from this limitation. However, the student misses the specific cause (the vanishing gradient problem) and does not mention RNNs or the LSTM solution involving memory cells and gating mechanisms. Thus, they capture the central symptom but omit key factual details from the Actual Answer.

Question 4

THEORY REMEMBER Best Score: 9/10

What is Masked Multi-Head Attention?

Desired Answer

Masked Multi-Head Attention is a special form of attention used in the **decoder of a Transformer model**. It ensures that when the model is predicting a word in a sequence, it can only attend to **previous words and not future ones**. This is achieved by applying a mask that blocks attention to positions corresponding to future tokens. The masking is necessary because during training the entire sentence is available, but during real generation the model must predict words **one step at a time** using only past information.

Student's Answer

Masked Multi-Head attention is used in decoder layer, it is typically used to avoid the model know about the next words which it has to predict.. For example: A cat sat on the mat.. lets say decoder is predicting the word after "the" , the input would only be till "the" ie., A cat sat on the ..., hiding the actual next word, so the model is not exposed to the answer..

Evaluations

Feedback 1

★★★★½ 9/10

The student correctly recalls that Masked Multi-Head Attention is used in the decoder and that it prevents the model from attending to future tokens by hiding the next word(s). The example illustrates the masking mechanism and the one-step prediction process. The only minor omission is the explicit mention of why masking is needed during training (because the whole sentence is available) versus during generation (predicting one token at a time), but all core facts from the Actual Answer are present and accurate.

Feedback 2

★★★★☆ 8/10

The student correctly states that masked multi-head attention is used in the decoder layer and that it prevents the model from seeing or attending to future words when predicting the next token (the core idea of blocking attention to future positions). They even give a clear example. However, they do not explicitly mention that this is in the context of a Transformer model, nor do they discuss the training versus real-generation rationale (that during training the full sentence is available but at inference you must predict one word at a time). These are secondary details in the Actual Answer that are omitted, so the recall is good but not complete.

Question 5

THEORY UNDERSTAND Best Score: 6/10

Why is cross-attention present in the decoder but not in the encoder?

Desired Answer

In a Transformer architecture, the encoder processes only the **input sequence**, so it relies on **self-attention** to understand relationships among the words in that input sentence. The decoder, on the other hand, generates the **output sequence** and therefore needs access to the information encoded by the encoder. Cross-attention allows the decoder to attend to the **encoder's output representations** while producing each token in the output. This mechanism helps the model align the input and output sequences, which is especially important in tasks such as machine translation.

Student's Answer

Student's Answer

Cross Attention is present in the decoder, it is used to attach encoder and decoder together, brings the output of encoder that is Key and Value, it then get processed in the decoder, in order to predict the output..

Evaluations

Feedback 1

★★★☆☆ 6/10

The student is addressing the same concepts as the Actual Answer (cross-attention in the decoder linking encoder outputs as key/value to predict outputs). Their answer shows a basic grasp of why cross-attention sits in the decoder (to bring in encoder representations for output generation). However, they omit the complementary point that the encoder only needs self-attention (because it only processes the input sequence), and they do not mention alignment or why cross-attention is crucial for tasks like translation. This is a partial but correct explanation of the core mechanism, meriting an Adequate Understanding score.

Feedback 2

★★★☆☆ 6/10

The student is addressing the correct concept—why cross-attention appears in the decoder—by explaining that it "attaches encoder and decoder together" and brings the encoder's outputs (as keys and values) into the decoder to predict the next token. This shows they understand that cross-attention lets the decoder use encoder representations. However, they do not mention why the encoder itself does not need cross-attention (i.e., it only processes the input and relies solely on self-attention), nor do they discuss how cross-attention helps align input and output sequences. Thus, their answer demonstrates a basic grasp of the decoder's cross-attention mechanism but misses the complementary explanation of the encoder's self-attention-only role and the alignment benefit. This partial but accurate coverage merits a score of 6/10.